



Department of Information Technology
Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CD3291 & Data Structures and Algorithms

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV / Tree Structures
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO11,12,13
- **Activity Chosen:** Think-Pair- Share
- **Justification:**

Tree is an important concept in data structures. Different types of tree structures were discussed in class and different list of values were assigned to various group of students. Some groups of students were instructed to build binary tree, some were instructed to build binary search tree, AVL tree, Heap and multiway search tree. Each student in groups was given a chance to excel themselves in a particular technique and share that knowledge with their team members. The main theme of this Think pair share activity is to promote both individual and peer learning which requires students to understand the concept very clearly and engage in discussion and learning.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - ✓ Instructor explained the concept of trees data structure in the previous class students were also instructed to go through the tree concepts.
 - ✓ Based on the discussion the teacher asked the students to think about the given list of elements and how to formulate it as a given category of tree 5 minutes individually without discussing with other in the class (Think). Next, pair up or make a group with others in the class and discuss their points with them 5 minutes and also listen their views (Pair).
 - ✓ Finally share points with the entire class (Share) as shown in Fig 1 & 2

- ✓ This assisted the students in recalling the topic taught on the previous week, generating new ideas about the topic, and answering questions about the topic with ease.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO4	3	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	2	2	2
Justification for correlation	Requires basic engineering knowledge to implement tree data structure	Able to identify engineering problems with tree data structures	Able to design solutions for complex engineering problems using tree data structure	Able to solve problem individually or in team and using tree data structures	Able to explain the way to solve the problem using tree with their team members.	Able to apply tree data structures to solve some computational problem.

- **Images / Screenshot of the practice:**



Figure 1: Students discussion with their team members [Naveen Kumar, Manish vardhan, Vignesh, Muniraj, Vijay Kumar]



Figure 2: Parasuram explains the solution of avl tree to other students in the class

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Every student was actively participated and done the task and practice among their group members.
 - Each student explored the knowledge of the tree data structure and it basic operations among the other students in the class.
 - It enables students to learn thoroughly about different types of tree structures.

- ❖ ***Benefit of the practice:***

- Students were actively participated in this activity.
 - From this activity, the students got clear idea about different tree data structures.
 - Slow learners are also able to perform implementation of tree data structure.

- ❖ ***Challenges faced in implementation:***

- It took some time and difficulties to organize the students into groups.
 - Some students in the team did not participate actively in the team.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GM_CS3353_Evaluation_Expression_TPS.pdf
 - ❖ <https://www.theteachertoolkit.com/index.php/tool/think-pair-share>
https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VI_Think_Pair_and_Share.pdf

Signature of Faculty Member

HOD